CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Basics of COA

Lecture No.- 01



By- Vishvadeep Gothi sir

Topics to be Covered



- Topic: Prerequisites
- Topic: Why COA
- Topic: Data In Computers
- Topic: Components of Computer
- Topic: Binary Numbers



Topic : Introduction

- **GATE Ranks:**
  - 682 (2009) – 3rd year
  - 19 (2010) – 4th year
  - 119, 440 etc.

Vishvadeep Gothi

- **Education:**
  - ME from IISc Bangalore
  - M. tech from BITS-pilani in Data Science (2018 - 20)

- **Work:**
  - 18+ Year Teaching Experience
  - 14+ in GATE/IES
  - Worked in Software Industry

Daily ⟶ Revision

chapter end ⟶ dpp (within 2 days)

weekend ⟶ Test (Compulsory)

PYQ ⟹ min. 2009 onwards

Basics of computers → Paid Course

Prerequisites of COA – 1
– 2
} youtube

@Vishvadeepsir



Topic : Prerequisites

- Basic components of computer: CPU, memory (RAM, ROM, HDD), I/O
- Number system: Binary, Decimal, Hexadecimal etc.
- Digital logic basics: Mux, Decoder etc.

Powers of 2:

Unit	Time	Bit or Byte
K (Kilo)	10^3	2^{10}
M (Mega)	10^6	2^{20}
G (Giga)	10^9	2^{30}
T (Tera)	10^{12}	2^{40}



Topic : Why COA

→ weightage ⇒ 8 - 10 Marks

↳ dynamic

- To understand: How a computer works
- To understand other courses: OS, Compiler, Programming etc.
- Help in real world development: DBMS, Hardware Design, IoT problems etc.



Topic : Computer Organization & Architecture

Computer Architecture:

- Conceptual design and fundamental operational structure.

CPU

memory

I/O

C.A.

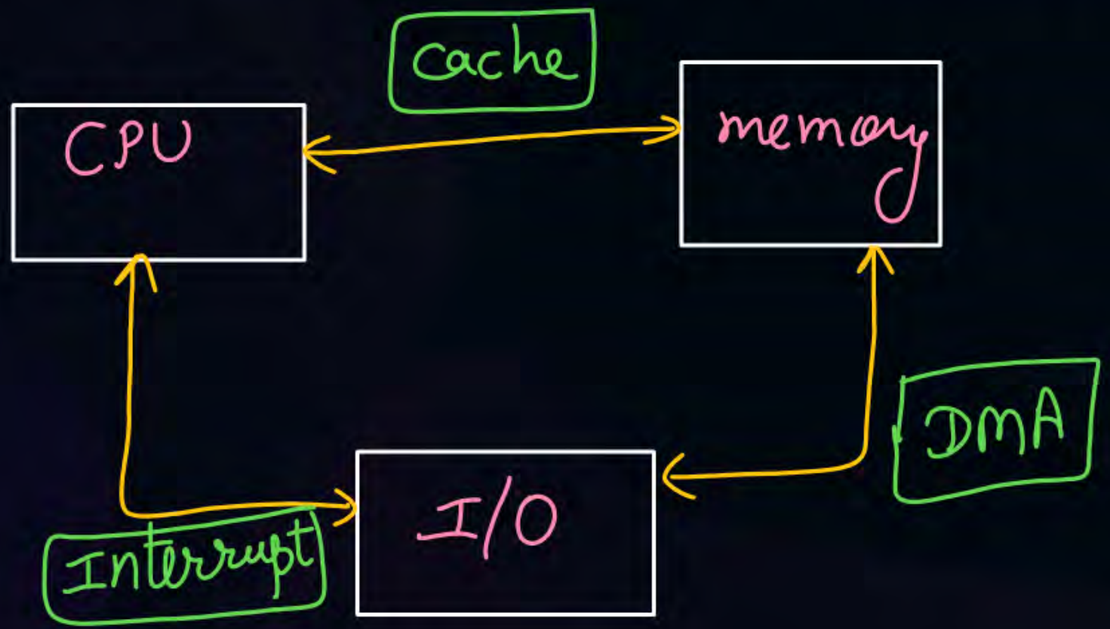
- CPU design
- Instructions
- Addressing modes
- Data format



Topic : Computer Organization & Architecture

Computer Organization: → implementation of architecture

- Deals with physical devices and their interconnections
- With a perspective of improving the performance.



C.O.
→ I/O organizatⁿ
→ mem. —11— —
→ Performance improvement



Topic : Computer Organization & Architecture

Computer Architecture	Computer Organization
• CPU Design	• I/O Organization
• Instructions	• Memory Organization
• Addressing modes	• Performance
• Data format	

In which format CPU will take data in binary.

→ Basics
→ micro operations and microprogramming
→ Instruction & Addressing modes
→ CPU & control Unit
→ Floating point
→ I/O organization
→ memory organization
→ Cache
→ Disk

→ Pipeline processing



Topic : Computer Organization & Architecture

Computer Architecture:

- Conceptual design and fundamental operational structure.

Computer Organization:

- Deals with physical devices and their interconnections
- With a perspective of improving the performance.

Computer Architecture	Computer Organization
• CPU Design	• I/O Organization
• Instructions	• Memory Organization
• Addressing modes	• Performance
• Data format	



Topic : Data In Computers

→ Processed and stored in binary format

- Data
  - Number
    - Fixed point
      - Unsigned
      - Signed
        - → 1's complement
        - → sign magnitude
        - → 2's complement
    - floating point
  - character
    - → ASCII (American standard code for information Interchang)
    - → EBCDIC (Extended Binary coded decimal interchang code)

ex:-

$5 = (101)_2$

$12 = (1100)_2$



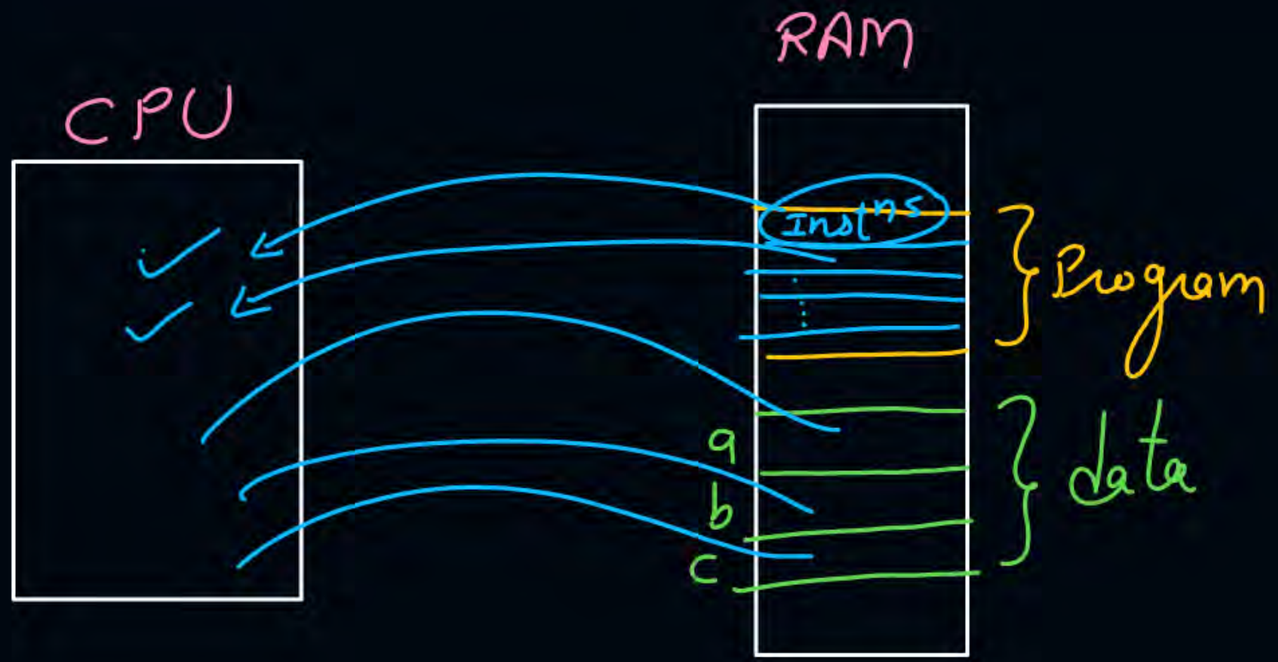
Topic : Components of Computer

- CPU (Parts) → Control Unit (C.U.)
  → Arithmetic Logic Unit (A.L.U.)

- Memory: (Primary/main)
  → RAM
  → ROM

- I/O Devices:
  → input devices
  → output devices
  → storage devices

How computer works :-




```
void main()
{
 int a, b, c;
 a = 5;
 b = 10;
 c = a + b;
 printf("%d", c);
}
```



Topic : Other Components

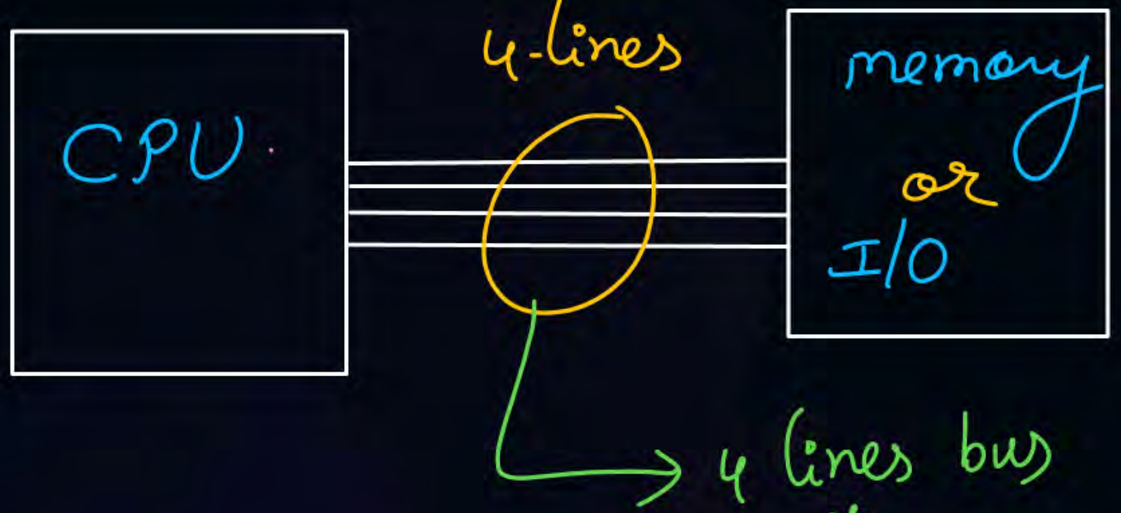
- System Buses
- CPU Registers



Topic : Other Components

8 - bits ⇒ 1 byte

System Buses: Collection of communication lines for components of computer



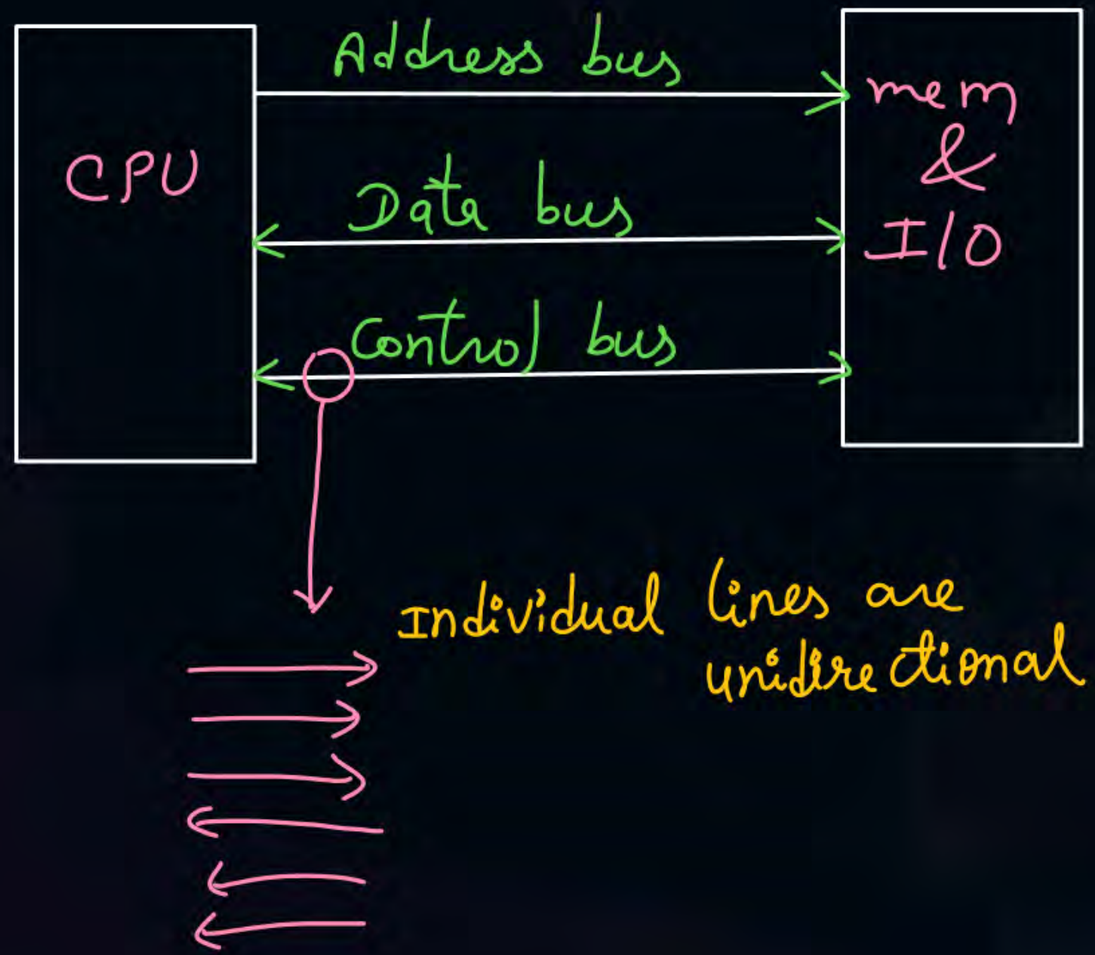
4 lines bus
or
4 bits bus
or
4 bits wide bus



Topic : Other Components

System Buses:

- Address Bus
- Data Bus
- Control Bus





2 mins Summary

- **Topic** Architecure vs Organization
- **Topic** Numbers & Data in Computers
- **Topic** Components of Computer
- **Topic** System Buses
- **Topic** Types of Buses

Happy Learning

THANK - YOU